

05 - Separation of variables

1. Use separation of variables to solve the ODE $\frac{dM}{dt} = 2.4 - 0.2M$.

2. Decide whether separation of variables will help you solve each ODE below. If you can, solve.

(a) $t \frac{dy}{dt} + 2y = t.$ (c) $\frac{dy}{dt} + 2ty = 0.$ (e) $y' + 2xy^2 = 0.$ (g) $\frac{dy}{dt} = \frac{t^2}{y^2(1+t^3)}.$

(b) $\frac{dy}{dt} = \frac{y}{\sin t}.$ (d) $\frac{dy}{dt} = \frac{\sin t}{y} + t.$ (f) $\frac{dy}{dt} = \frac{t}{3y^2} + 3$ (h) $\frac{dy}{dt} = \frac{6y-2}{ty}.$

3. Solve $y' = x - xy$ using separation of variables.

4. Solve the differential equation $\frac{dy}{dt} = -\frac{t}{y}$ using separation of variables.

5. **Solving the logistic equation.** Remember the logistic equation from last time? Solve it in this particular case: $y' = 2y(5 - y)$. *Hint: you might need to review partial fractions.*

Can you also (or instead) solve it in more generality, that is, solve $y' = ky(L - y)$?

6. Solve the ODE $P' = P(1 - P)(P - 2)$. You should be able to find an explicit solution.

In problem 9 from last worksheet we saw the ODE $P' = \frac{P}{2} \left(1 - \frac{P}{5}\right) \left(\frac{P}{8} - 1\right)$, which was used to make predictions about the spotted owl population in a forest in the Canadian Rockies. Even though this ODE is qualitatively very similar to $P' = P(1 - P)(P - 2)$, you will not be able to solve explicitly the owl ODE. Explain why.

7. **Daily Problem. Please try this question in preparation for next class.**

- (a) Solve the differential equation $\frac{d}{dt}(ty) = 5t$ for the unknown function $y(t)$.
- (b) Does your work help you solve $ty' + y = 5t$ for $y(t)$?
- (c) Also, does your work now help you solve $y' + \frac{1}{t}y = 5$ for $y(t)$?